


```
# make
```

3. Start wpa_supplicant

- i. Run wpa_supplicant in background:

If cfg80211:

```
# wpa_supplicant -Dnl80211 -iwlan0 -c ../wpa_0_8.conf B
```

Or wireless extensions:

```
# wpa_supplicant -Dwext -iwlan0 -c ../wpa_0_8.conf -B
```

- ii. Run wpa_supplicant in background with debug message (This may affect the performance, only used in debug purpose.)

If cfg80211:

```
# wpa_supplicant -Dnl80211 -iwlan0 -c ../wpa_0_8.conf -dd &
```

Or wireless extension:

```
# wpa_supplicant -Dwext -iwlan0 -c ../wpa_0_8.conf -dd &
```

(B) WPA_CLI commands

1. Scanning AP and See Results

```
# wpa_cli -p/var/run/wpa_supplicant scan  
# wpa_cli -p/var/run/wpa_supplicant scan_results
```

2. Connect to AP

- a. OPEN

```
# wpa_cli -p/var/run/wpa_supplicant remove_network 0  
# wpa_cli -p/var/run/wpa_supplicant ap_scan 1  
# wpa_cli -p/var/run/wpa_supplicant add_network  
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ssid ""dlink""  
# wpa_cli -p/var/run/wpa_supplicant set_network 0 key_mgmt NONE  
# wpa_cli -p/var/run/wpa_supplicant select_network 0
```

- b. WEP40 with open system

```
# wpa_cli -p/var/run/wpa_supplicant remove_network 0  
# wpa_cli -p/var/run/wpa_supplicant ap_scan 1  
# wpa_cli -p/var/run/wpa_supplicant add_network  
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ssid ""dlink""  
# wpa_cli -p/var/run/wpa_supplicant set_network 0 key_mgmt NONE  
# wpa_cli -p/var/run/wpa_supplicant set_network 0 wep_key0 1234567890  
# wpa_cli -p/var/run/wpa_supplicant set_network 0 wep_tx_keyidx 0  
# wpa_cli -p/var/run/wpa_supplicant select_network 0
```

- c. WEP40 with shared key mode

```
# wpa_cli -p/var/run/wpa_supplicant remove_network 0  
# wpa_cli -p/var/run/wpa_supplicant ap_scan 1  
# wpa_cli -p/var/run/wpa_supplicant add_network  
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ssid ""dlink""
```

wpa_cli

Realtek

```
# wpa_cli -p/var/run/wpa_supplicant set_network 0 key_mgmt SAE
# wpa_cli -p/var/run/wpa_supplicant set_network 0 psk ""12345678""
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ieee80211w 2
# wpa_cli -p/var/run/wpa_supplicant select_network 0
```

h. WPA3-SAE Transition Mode (MFPC=1, MFPR=0)

```
# wpa_cli -p/var/run/wpa_supplicant remove_network 0
# wpa_cli -p/var/run/wpa_supplicant ap_scan 1
# wpa_cli -p/var/run/wpa_supplicant add_network
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ssid ""dlink""
# wpa_cli -p/var/run/wpa_supplicant set_network 0 key_mgmt SAE WPA-PSK
# wpa_cli -p/var/run/wpa_supplicant set_network 0 psk ""12345678""
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ieee80211w 1
# wpa_cli -p/var/run/wpa_supplicant select_network 0
```

3. Ad-hoc mode

a. OPEN

```
# wpa_cli -p/var/run/wpa_supplicant scan
# wpa_cli -p/var/run/wpa_supplicant scan_results
# wpa_cli -p/var/run/wpa_supplicant remove_network 0
# wpa_cli -p/var/run/wpa_supplicant ap_scan 2
# wpa_cli -p/var/run/wpa_supplicant add_network
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ssid ""Adhoc_test""
# wpa_cli -p/var/run/wpa_supplicant set_network 0 mode 1
# wpa_cli -p/var/run/wpa_supplicant set_network 0 key_mgmt NONE
# wpa_cli -p/var/run/wpa_supplicant set_network 0 frequency 2412
# wpa_cli -p/var/run/wpa_supplicant select_network 0
```

#frequency is to set the channel frequency for Ad-hoc master.

b. WEP40

```
# wpa_cli -p/var/run/wpa_supplicant scan
# wpa_cli -p/var/run/wpa_supplicant scan_results
# wpa_cli -p/var/run/wpa_supplicant remove_network 0
# wpa_cli -p/var/run/wpa_supplicant ap_scan 2
# wpa_cli -p/var/run/wpa_supplicant add_network
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ssid ""Adhoc_test""
# wpa_cli -p/var/run/wpa_supplicant set_network 0 mode 1
# wpa_cli -p/var/run/wpa_supplicant set_network 0 key_mgmt NONE
# wpa_cli -p/var/run/wpa_supplicant set_network 0 wep_key0 1234567890
# wpa_cli -p/var/run/wpa_supplicant set_network 0 wep_tx_keyidx 0
# wpa_cli -p/var/run/wpa_supplicant set_network 0 frequency 2412
# wpa_cli -p/var/run/wpa_supplicant select_network 0
```

c. WEP104

```
# wpa_cli -p/var/run/wpa_supplicant scan
# wpa_cli -p/var/run/wpa_supplicant scan_results
# wpa_cli -p/var/run/wpa_supplicant remove_network 0
# wpa_cli -p/var/run/wpa_supplicant ap_scan 2
# wpa_cli -p/var/run/wpa_supplicant add_network
# wpa_cli -p/var/run/wpa_supplicant set_network 0 ssid ""Adhoc_test""
# wpa_cli -p/var/run/wpa_supplicant set_network 0 mode 1
# wpa_cli -p/var/run/wpa_supplicant set_network 0 key_mgmt NONE
# wpa_cli -p/var/run/wpa_supplicant set_network 0 wep_key0
```

```
12345678901234567890123456
# wpa_cli -p/var/run/wpa_supplicant set_network 0 wep_tx_keyidx 0
# wpa_cli -p/var/run/wpa_supplicant set_network 0 frequency 2412
# wpa_cli -p/var/run/wpa_supplicant select_network 0
```

4. Save the Current Connection AP configuration file

```
# wpa_cli -p/var/run/wpa_supplicant save_config
```

5. WPS Connection

(1) Push Button:

```
# wpa_cli -p/var/run/wpa_supplicant remove_network 0
# wpa_cli -p/var/run/wpa_supplicant wps_pbc any
```

(2) Pin Code:

```
# wpa_cli -p/var/run/wpa_supplicant remove_network 0
# wpa_cli -p/var/run/wpa_supplicant wps_pin any 12345670
```

Or

```
# wpa_cli -p/var/run/wpa_supplicant remove_network 0
# wpa_cli -p/var/run/wpa_supplicant wps_pin any
```

6. Get Current Status of wpa_supplicant

```
# wpa_cli -p/var/run/wpa_supplicant status
```

7. Disable current network connection

```
# wpa_cli -p/var/run/wpa_supplicant disable_network 0
```

(C) Using WPA_SUPPLICANT by WPA_CLI (Control interface commands)

1. Start **wpa_cli** control interface:

```
# wpa_cli
```

2. Commands:

PING

This command can be used to test whether wpa_supplicant is replying to the control interface commands.

The expected reply is PONG if the connection is open and wpa_supplicant is processing commands.

STATUS

Request current status information. The output is a text block with each line in variable=value format. For example:

bssid=02:00:01:02:03:04
ssid=test network
pairwise_cipher=CCMP
group_cipher=CCMP
key_mgmt=WPA-PSK
wpa_state=COMPLETED

LIST_NETWORKS

List configured networks.

network id / ssid / bssid / flags

0 example network any [CURRENT]

(note: fields are separated with tabs)

SCAN

Request a new BSS scan.

SCAN_RESULTS

Get the latest scan results.

bssid / frequency / signal level / flags / ssid

00:09:5b:95:e0:4e 2412 208 [WPA-PSK-CCMP] jkm private

02:55:24:33:77:a3 2462 187 [WPA-PSK-TKIP] testing

00:09:5b:95:e0:4f 2412 209 jkm guest

(note: fields are separated with tabs)

ADD_NETWORK

Add a new network. This command creates a new network with empty configuration. The new network is

disabled and once it has been configured it can be enabled with ENABLE_NETWORK command.

ADD_ -

NETWORK returns the [network id](#) of the new network or FAIL on failure

SELECT_NETWORK <network id>

Select a network (disable others). Network id can be received from the LIST_NETWORKS command output.

ENABLE_NETWORK <network id>

Enable a network. Network id can be received from the LIST_NETWORKS command output.

DISABLE_NETWORK <network id>

Disable a network. Network id can be received from the LIST_NETWORKS command output.

Special

network id [all](#) can be used to disable all network.

.

REMOVE_NETWORK <network id>

Remove a network. Network id can be received from the LIST_NETWORKS command output.

Special

network id [all](#) can be used to remove all network.

SET_NETWORK <network id> <variable> <value>

Set network variables. Network id can be received from the LIST_NETWORKS command output. This command uses the same variables and data formats as the configuration file.

```
! ! ) ! ! E*
! ! ) B! ! ! -shared key)
! ! ! ) ! ! ! F! B-PSK, WPA-EAP)
! ! ! ) B! B3*
! ! ! ) DD ! L *
! ! ! ) DD ! L ! F 51! F 215*
! ! ! ! ! ! ! ! 1*
! ! ! ! ! ! ! ! *
!frequency ( Channel frequency in megahertz (MHz) for IBSS )
```

GET_NETWORK <network id> <variable>

Get network variables. Network id can be received from the LIST_NETWORKS command output.

SAVE_CONFIG

Save the current configuration.

AP_SCAN <ap_scan value>

Change ap_scan value: 0 = no scanning, 1 = wpa_supplicant requests scans and uses scan results to select the AP, 2 = wpa_supplicant does not use scanning and just requests driver to associate and take care of AP selection